

Skewness calculation.

data = [12, 15, 14, 10, 18, 20, 22, 25, 13, 17, 19, 21, 16, 14, 23]

Ascending order:

10, 12, 13, 14, 14, 15, 16, 17, 18, 19, 20, 21, 23, 25.

$n = 15$.

$Med = 17$ (8th)

* Mean

$$\rightarrow \frac{12+15+14+10+18+20+22+25+13+17+19+21+16+14+23}{15}$$

$$\bar{X} = 17.27$$

* Standard Deviation:

$$\sigma = \sqrt{\frac{1}{n} \cdot \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \sqrt{\frac{1}{15} \left((10-17.27)^2 + (12-17.27)^2 + (13-17.27)^2 + (14-17.27)^2 + (14-17.27)^2 + (15-17.27)^2 + (16-17.27)^2 + (17-17.27)^2 + (18-17.27)^2 + (19-17.27)^2 + (20-17.27)^2 + (21-17.27)^2 + (16-17.27)^2 + (14-17.27)^2 + (23-17.27)^2 + (25-17.27)^2 \right)}$$

$$= 4.16$$

As we know,

$$\text{Skewness} = \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

$$= \frac{3(17.27 - 17)}{4.16} = 0.19$$

$$\text{data} = [5, 7, 8, 9, 10, 12, 15, 18, 22, 30, 45, 60]$$

$$n = 12$$

Mean

$$\bar{x} = \frac{(5+7+8+9+10+12+15+18+22+30+45+60)}{12}$$
$$= 20.08$$

$$Md = \frac{12+15}{2} = 13.5$$

$$S.D (\sigma) = \sqrt{\frac{\sum (x - \bar{x})^2}{12}}$$

$$= \sqrt{\frac{1}{12} \left[(5-20.08)^2 + (7-20.08)^2 + (8-20.08)^2 + (9-20.08)^2 + \right.}$$
$$\left. + (10-20.08)^2 + (12-20.08)^2 + (15-20.08)^2 + \right.}$$
$$\left. (18-20.08)^2 + (22-20.08)^2 + (30-20.08)^2 + \right.}$$
$$\left. (45-20.08)^2 + (60-20.08)^2 \right]$$

$$= 16.6$$

$$\text{Skewness} = \frac{3(\bar{x} - Md)}{\sigma}$$
$$= \frac{3(20.08 - 13.5)}{16.6}$$
$$= 1.22$$

Pen: kurtosis formula.

$$\beta_2 = \frac{\sum x_i^4}{\sum x_i^2}$$

$a = [4.2, 5.1, 4.8, 5.5, 4.9, 5.0, 5.3, 4.7, 5.2, 4.6, 5.4, 4.5, 5.1, 4.9, 5.0, 5.2, 4.8, 5.3, 4.7, 5.1]$.

$$\bar{a} = \frac{\sum_{i=1}^{20} a_i}{20} \quad \text{Here } n=20.$$

$$= \frac{99.8}{20} = 4.99$$

Taking first 10 element as a sample size.

$$\bar{a} = \frac{\sum_{i=1}^{10} a_i}{10} = \frac{49.8}{10} = 4.98$$

Deviation Table.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^4$
4.2	-0.78	0.5329	0.283982
5.1	0.17	0.0289	0.000835
4.8	-0.18	0.0169	0.000286
5.5	0.57	0.3249	0.105560
4.9	-0.03	0.0009	0.000001
5.0	0.07	0.0049	0.000024
5.3	0.37	0.1369	0.018742
4.7	-0.23	0.0529	0.002798
5.2	0.27	0.0729	0.005314
4.6	-0.33	0.1089	0.011859
		$\sum (x_i - \bar{x})^2$	$\sum (x_i - \bar{x})^4$
		1.2810	= 0.429402

sd

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{1.2810}{9}}$$

$$= 0.3772$$

Now,

$$m_4 = \frac{1}{n} \sum (x_i - \bar{x})^4$$

$$= \frac{1}{10} \times 0.429402 = 0.042940$$

$$m_2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{1.2810}{10} = 0.12810$$

$$\beta_2 = \frac{m_4}{(m_2)^2} = \frac{0.042940}{(0.12810)^2} = 2.6168$$

$\beta_2 < 3 \Rightarrow$ platykurtic.

Excess kurtosis

$$\hookrightarrow k - 3$$

$$= 2.61 - 3$$

$$= -0.61$$

$C = [3, 0, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 3.2, 3.8, 4.2, 4.8, 5.2, 5.8, 6.2, 6.8, 7.2]$

$$\bar{C} = \frac{\sum_{i=1}^{18} C}{N} = \frac{92.2}{18} = 5.1222$$

$C_i - \bar{C}$	$(C_i - \bar{C})^2$	$(C_i - \bar{C})^4$
-2.1222	4.5038	20.280
-1.6222	2.6315	6.925
-1.1222	1.2593	1.586
-0.6222	0.3871	0.150
-0.1222	0.0149	0.00022
0.3778	0.1427	0.0209
0.8778	0.7705	0.594
1.3778	1.8983	3.604
1.8778	3.5267	12.440
-1.9222	3.6959	13.655
-1.3222	1.7482	3.055
-0.9222	0.8505	0.727
-0.3222	0.1038	0.011
0.10778	0.0061	0.000037
0.6778	0.4594	0.211
1.0778	1.1616	1.349
1.6778	2.8153	7.927
2.0778	4.3169	18.645
	$= 30.0424$	90.146

$$\begin{aligned} \text{kurtosis} &= \frac{\mu_4}{(\sigma_2)^2} = \frac{5008}{(1.668)^2} \\ &= \frac{5.008}{2.783} \\ &= 1.799 \end{aligned}$$

Calculate excess kurtosis.

$$\begin{aligned} \text{Kurtosis} &- 3 \\ &= 1.799 - 3 \\ &= -1.201 \end{aligned}$$

Platykurtic.

Now, $\sigma_2 = \frac{30.0424}{18} = 1.668$

$$\mu_4 = \frac{90.146}{18} = 5.008$$